Search environments

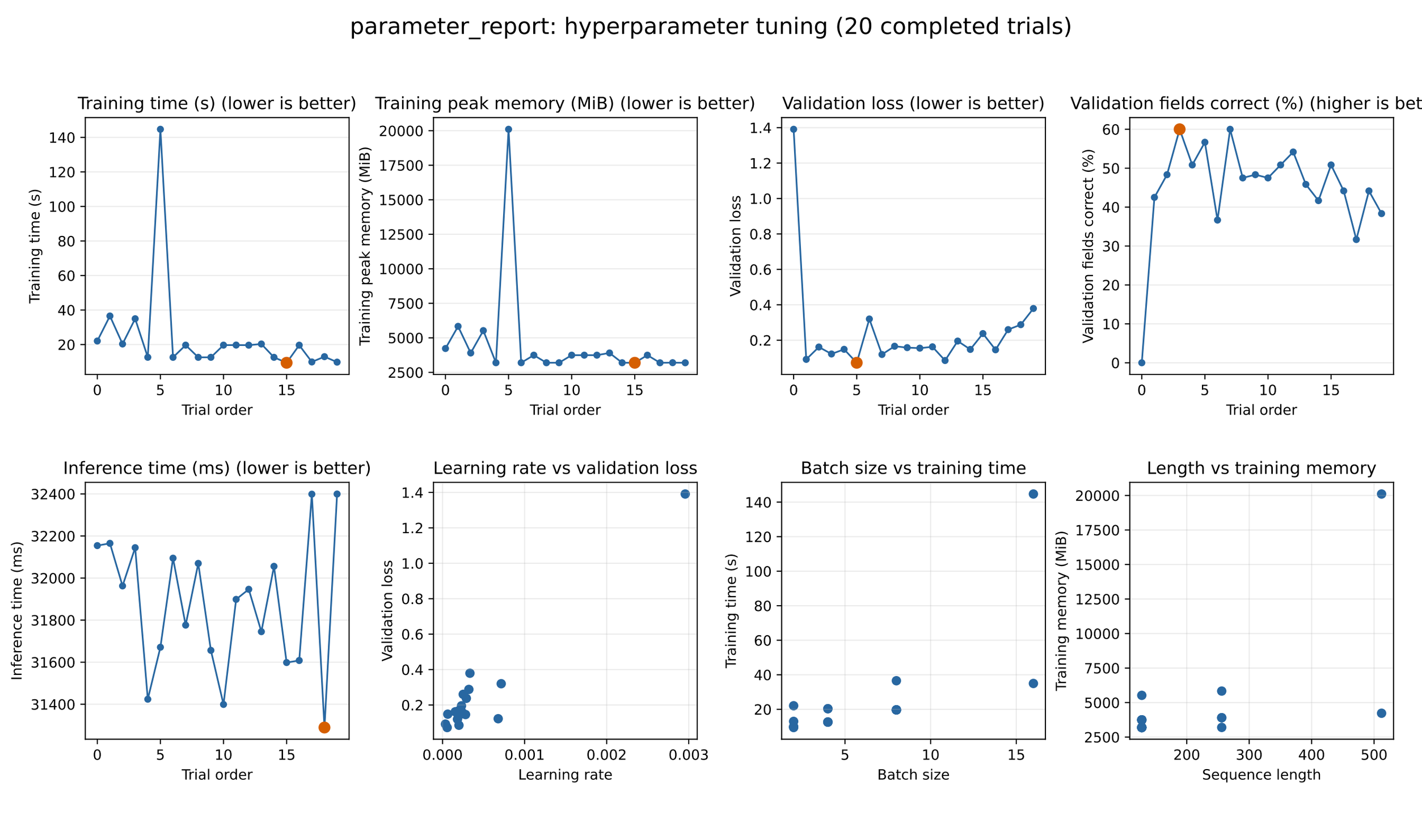
parameter_report
Python 3.12.9; PyTorch 2.14.0+cu130; CUDA 13.0; GPU NVIDIA A100-SXM4-80GB
Platform: Linux-6.8.0-124-generic-x86_64-with-glibc2.39; executable: /home/ovhart/.conda/envs/MQP/bin/python

parameter_report_flash (flash_attention)
Python 3.12.9; PyTorch 2.14.0+cu130; CUDA 13.0; GPU NVIDIA A100-SXM4-80GB
Platform: Linux-6.8.0-124-generic-x86_64-with-glibc2.39; executable: /home/ovhart/.conda/envs/MQP/bin/python

parameter_report_kvcache (kvcache)
Python 3.12.9; PyTorch 2.14.0+cu130; CUDA 13.0; GPU NVIDIA A100-SXM4-80GB
Platform: Linux-6.8.0-124-generic-x86_64-with-glibc2.39; executable: /home/ovhart/.conda/envs/MQP/bin/python

parameter_report_lora (lora)
Python 3.12.9; PyTorch 2.14.0+cu130; CUDA 13.0; GPU NVIDIA A100-SXM4-80GB
Platform: Linux-6.8.0-124-generic-x86_64-with-glibc2.39; executable: /home/ovhart/.conda/envs/MQP/bin/python

parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention)
Python 3.12.9; PyTorch 2.14.0+cu130; CUDA 13.0; GPU NVIDIA A100-SXM4-80GB
Platform: Linux-6.8.0-124-generic-x86_64-with-glibc2.39; executable: /home/ovhart/.conda/envs/MQP/bin/python

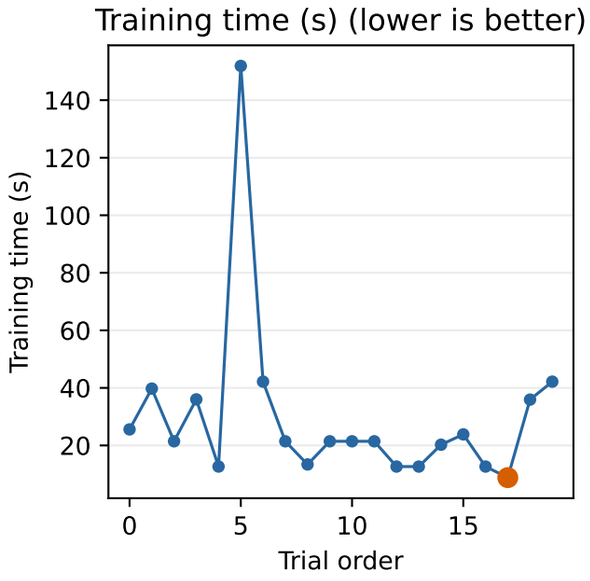

parameter_report: hyperparameter tuning (20 completed trials)
Training time (s) (lower is better)
Training time (s)
Trial order
Training peak memory (MiB) (lower is better)
Training peak memory (MiB)
Trial order
Validation loss (lower is better)
Validation loss
Trial order
Validation fields correct (%) (higher is bet
Validation fields correct (%)
Trial order
Inference time (ms) (lower is better)
Inference time (ms)
Trial order
Learning rate vs validation loss
Validation loss
Learning rate
Batch size vs training time
Training time (s)
Batch size
Length vs training memory
Training memory (MiB)
Sequence length

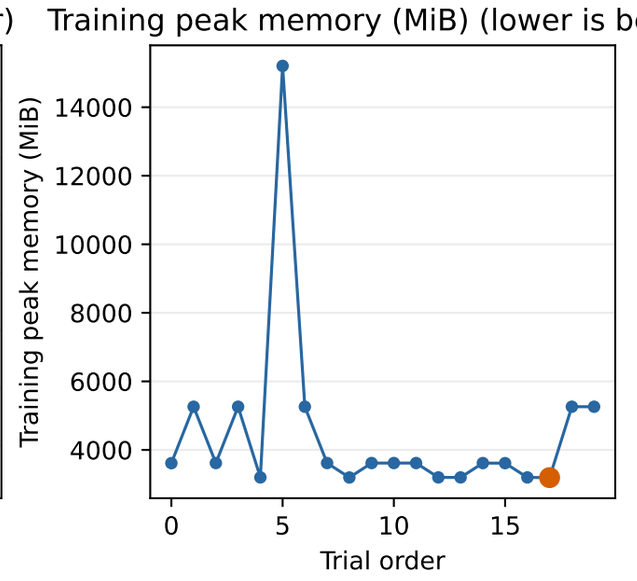
parameter_report: trial hyperparameters

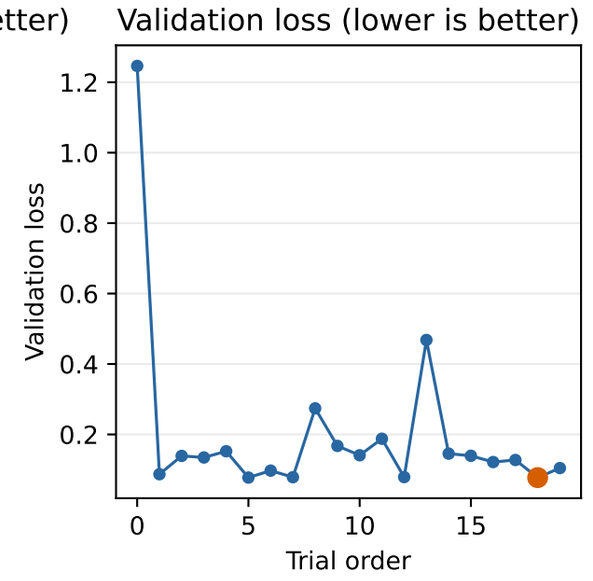
Trial	Learning Rate	Batch Size	Length
0	0.0029578	2	512
1	3.5613e-05	8	256
2	0.00015536	4	256
3	0.00067814	16	128
4	**0.00022618**	**4**	**128**
5	5.6412e-05	16	512
6	0.00071498	4	128
7	0.00018216	8	128
8	0.00019167	4	128
9	0.00023868	4	128
10	0.00021795	8	128
11	0.00022683	8	128
12	0.0002	8	128
13	0.00023092	4	256
14	6.3543e-05	4	128
15	0.00028977	2	128
16	0.00028011	8	128
17	0.00025203	2	128
18	0.00032064	2	256
19	0.0003337	2	128

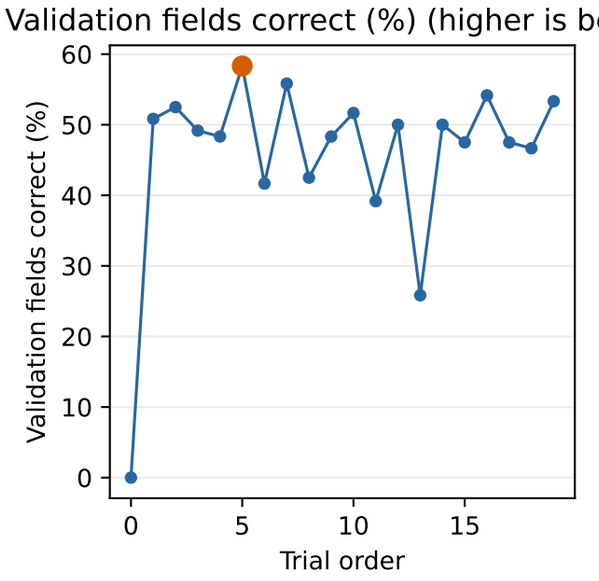
Highlighted row: selected best trial based on this report's objectives.

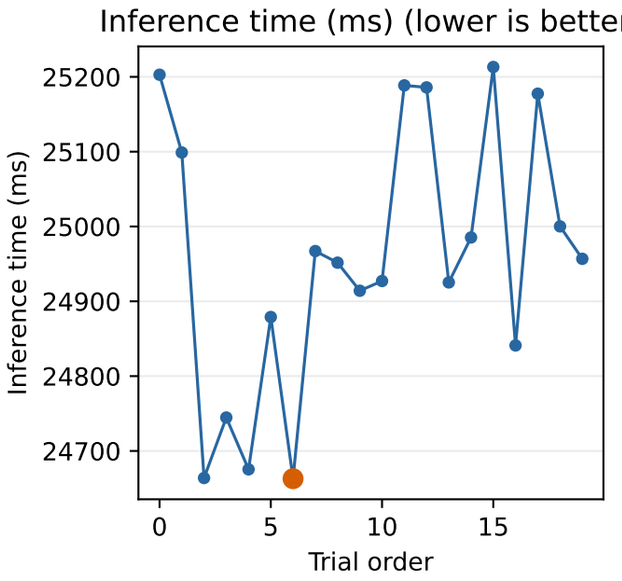
parameter_report_flash (flash_attention): hyperparameter tuning (20 completed trials)

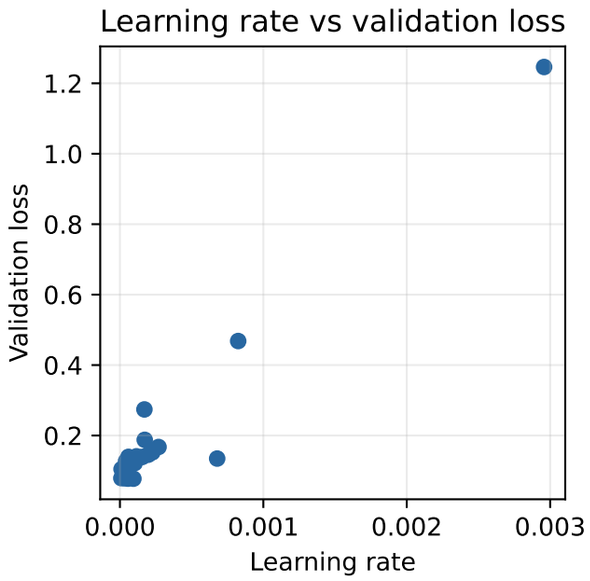


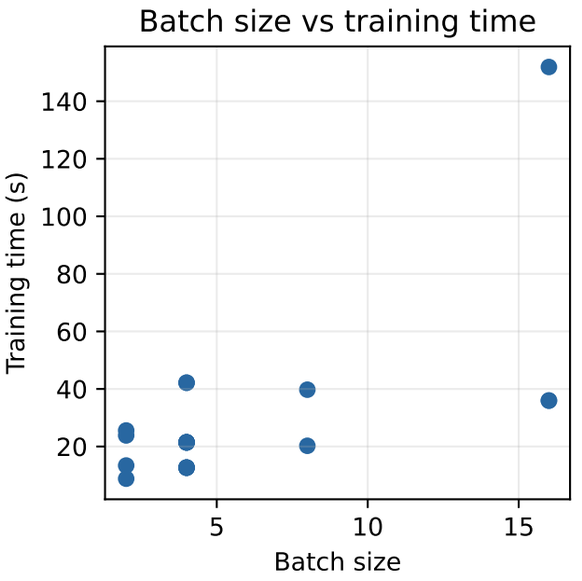


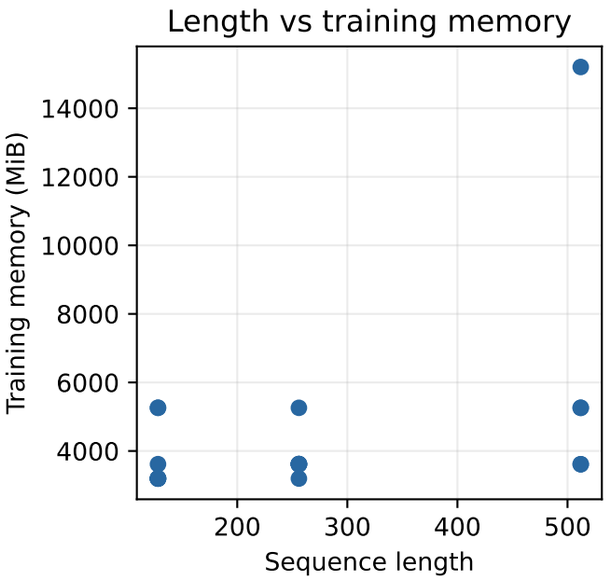








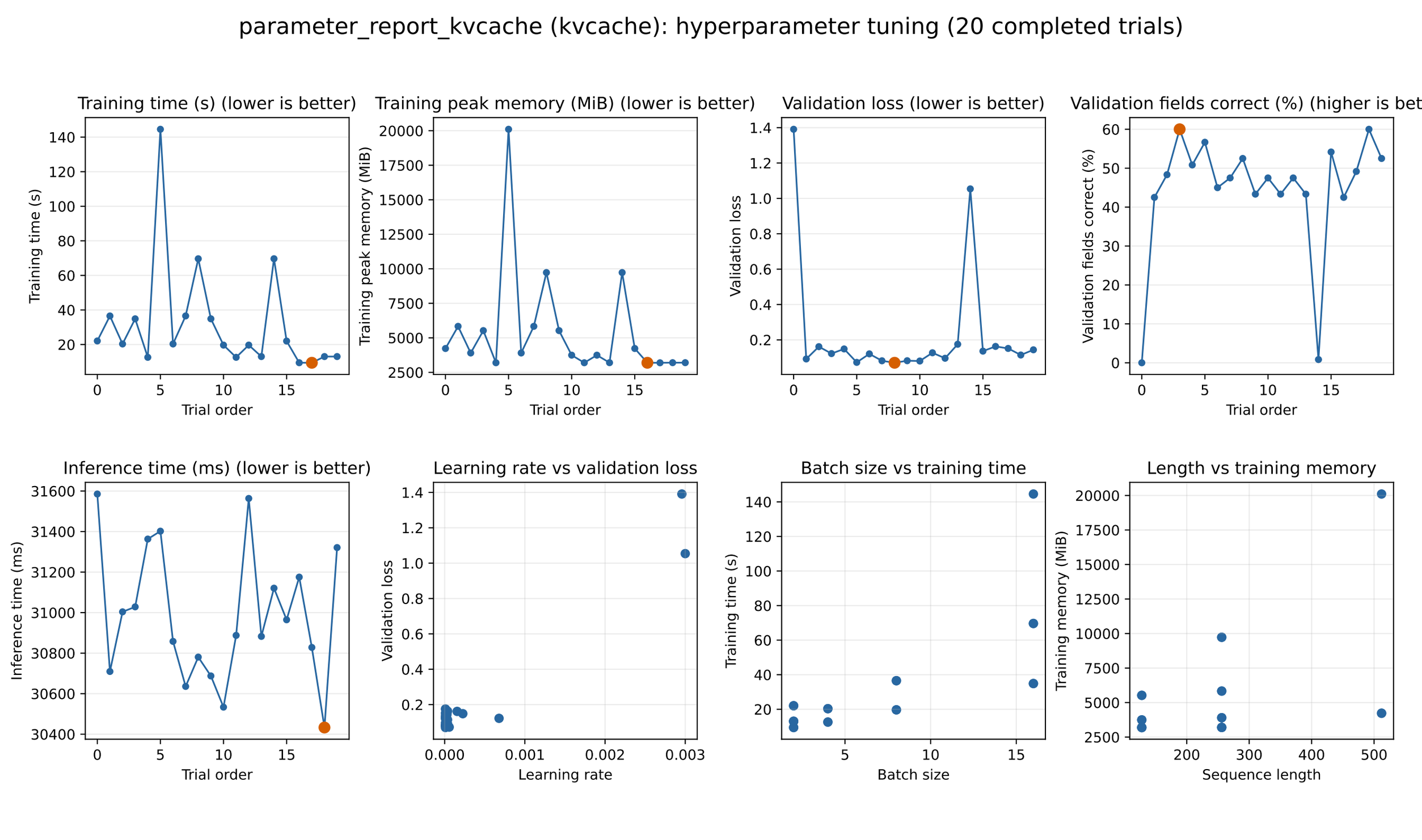




parameter_report_flash (flash_attention): trial hyperparameters

Trial	Learning Rate	Batch Size	Length
0	0.0029578	2	512
1	3.5613e-05	8	256
2	**0.00015536**	**4**	**256**
3	0.00067814	16	128
4	0.00022618	4	128
5	5.6412e-05	16	512
6	3.651e-05	4	512
7	3.4268e-05	4	256
8	0.00017066	2	256
9	0.00026922	4	256
10	0.00011515	4	256
11	0.0001728	4	256
12	1e-05	4	128
13	0.00082462	4	128
14	0.00019751	8	128
15	5.9703e-05	2	512
16	0.00010282	4	128
17	4.2832e-05	2	128
18	9.3996e-05	16	128
19	1.1381e-05	4	512

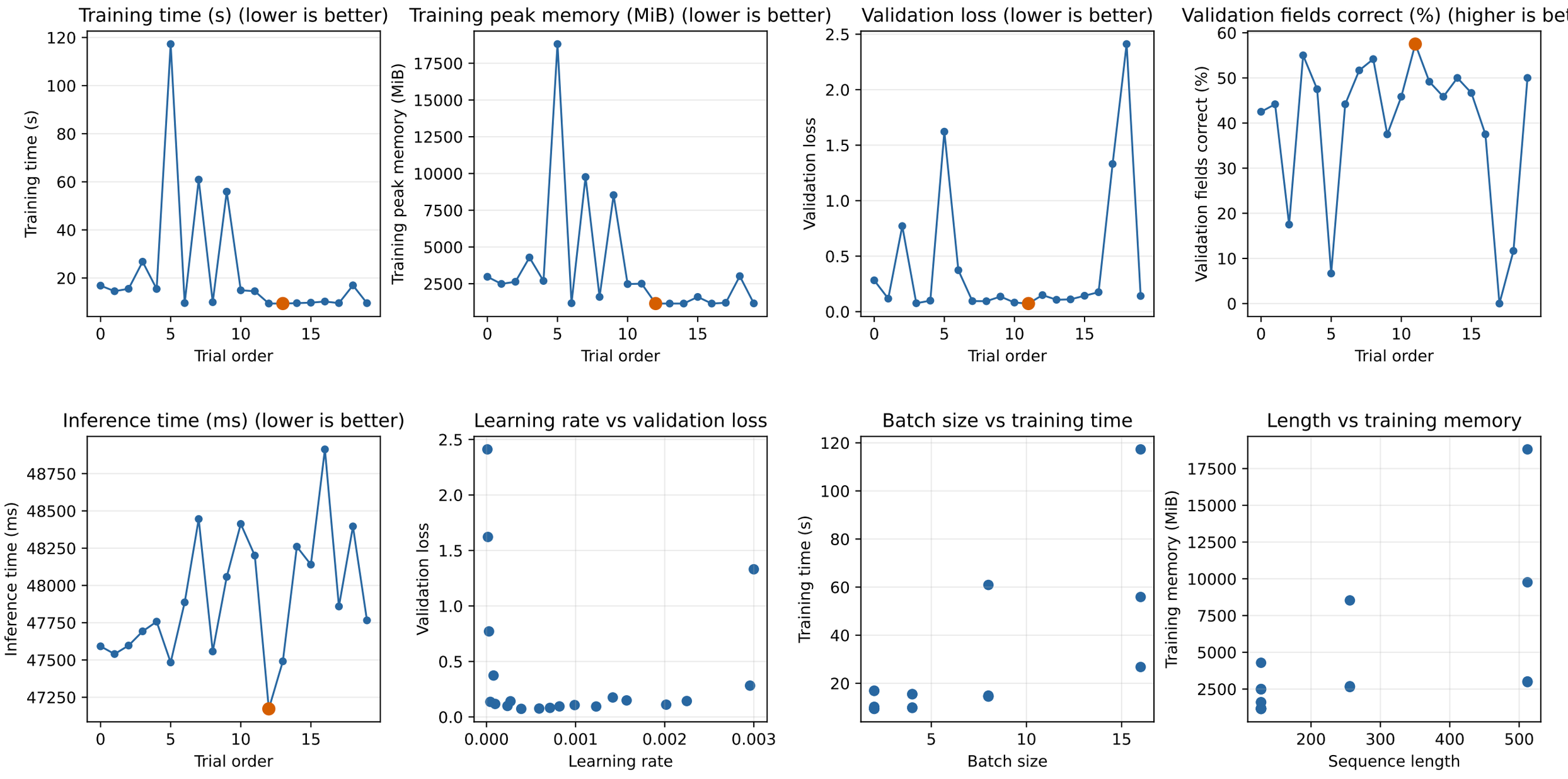
Highlighted row: selected best trial based on this report's objectives.


parameter_report_kvcache (kvcache): hyperparameter tuning (20 completed trials)
Training time (s) (lower is better)
Training time (s)
Trial order
Training peak memory (MiB) (lower is better)
Training peak memory (MiB)
Trial order
Validation loss (lower is better)
Validation loss
Trial order
Validation fields correct (%) (higher is bet
Validation fields correct (%)
Trial order
Inference time (ms) (lower is better)
Inference time (ms)
Trial order
Learning rate vs validation loss
Validation loss
Learning rate
Batch size vs training time
Training time (s)
Batch size
Length vs training memory
Training memory (MiB)
Sequence length

parameter_report_kvcache (kvcache): trial hyperparameters

Trial	Learning Rate	Batch Size	Length
0	0.0029578	2	512
1	3.5613e-05	8	256
2	0.00015536	4	256
3	0.00067814	16	128
4	0.00022618	4	128
5	5.6412e-05	16	512
6	1e-05	4	256
7	1e-05	8	256
8	1e-05	16	256
9	1e-05	16	128
10	1e-05	8	128
11	1e-05	4	128
12	1e-05	8	128
13	1e-05	2	256
14	0.003	16	256
15	1e-05	2	512
16	3.5694e-05	2	128
17	1e-05	2	128
18	**3.5742e-05**	**2**	**256**
19	2.8131e-05	2	256

Highlighted row: selected best trial based on this report's objectives.

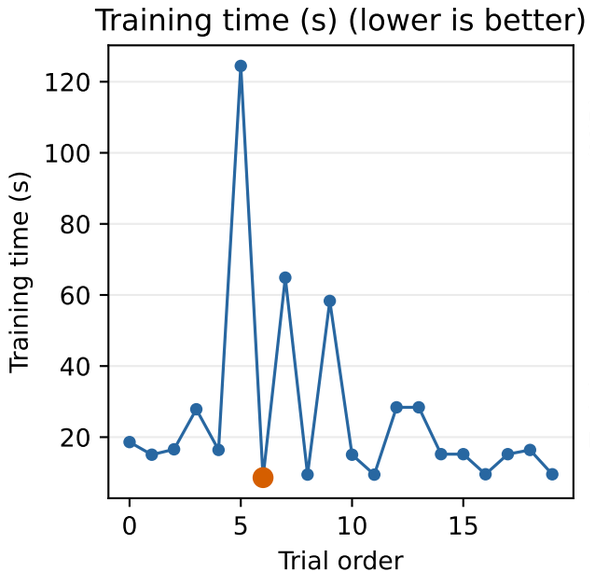

parameter_report_lora (lora): hyperparameter tuning (20 completed trials)
Training time (s) (lower is better)
Training time (s)
Trial order
Training peak memory (MiB) (lower is better)
Training peak memory (MiB)
Trial order
Validation loss (lower is better)
Validation loss
Trial order
Validation fields correct (%) (higher is bet
Validation fields correct (%)
Trial order
Inference time (ms) (lower is better)
Inference time (ms)
Trial order
Learning rate vs validation loss
Validation loss
Learning rate
Batch size vs training time
Training time (s)
Batch size
Length vs training memory
Training memory (MiB)
Sequence length

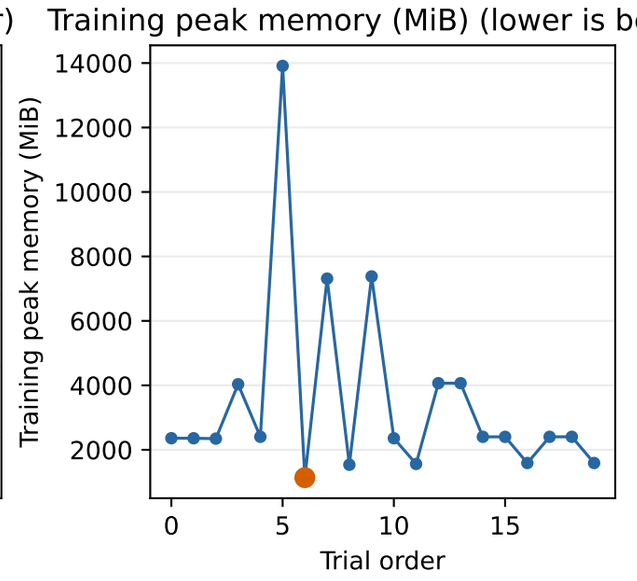
parameter_report_lora (lora): trial hyperparameters

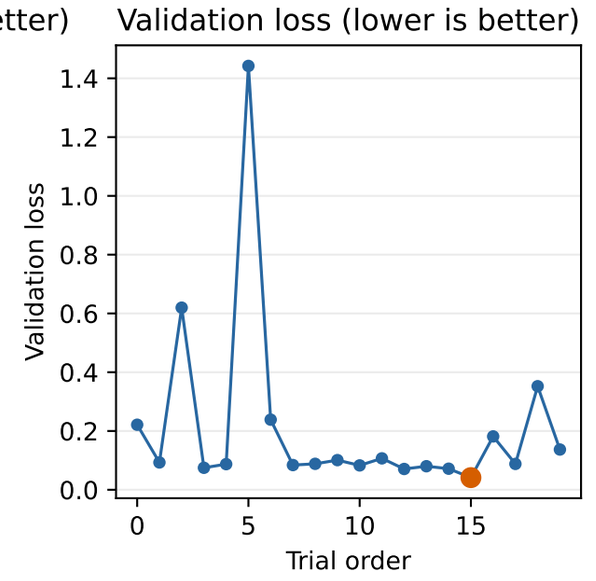
Trial	Learning Rate	Batch Size	Length	Rank	Alpha
0	0.0029578	2	512	8	16
1	9.7448e-05	8	128	8	16
2	2.807e-05	4	256	2	8
3	0.00059155	16	128	16	32
4	0.00023533	4	256	32	64
5	1.5861e-05	16	512	2	4
6	7.9071e-05	2	128	16	8
7	0.00081909	8	512	4	32
8	0.0012308	4	128	2	8
9	4.2976e-05	16	256	32	32
10	0.0007125	8	128	2	8
11	0.00039004	8	128	16	16
12	**0.0015726**	**2**	**128**	**2**	**8**
13	0.00098932	2	128	2	8
14	0.0020167	2	128	2	4
15	0.0022479	4	128	2	8
16	0.0014173	2	128	2	8
17	0.003	2	128	32	64
18	1e-05	2	512	32	4
19	0.00026929	2	128	8	64

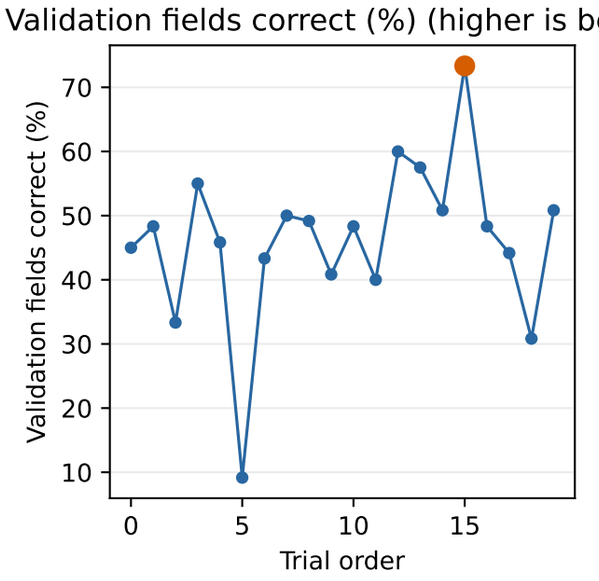
Highlighted row: selected best trial based on this report's objectives.

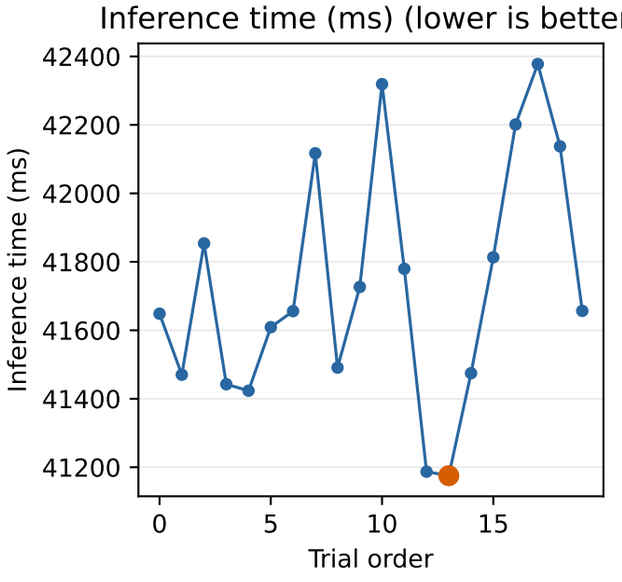
parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention): hyperparameter tuning (20 completed trials)

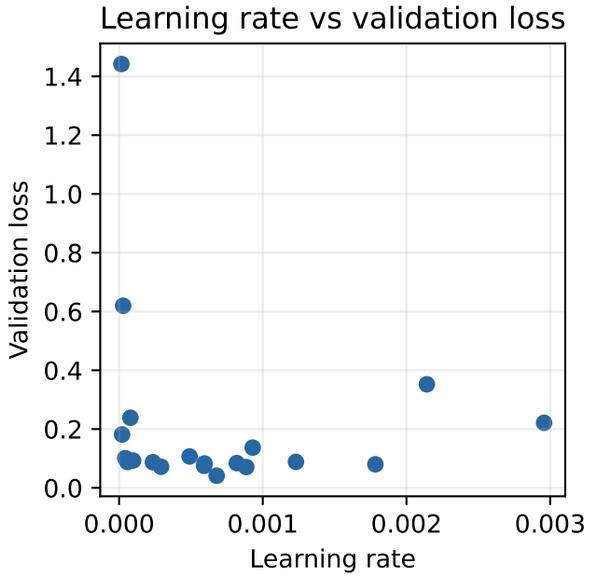


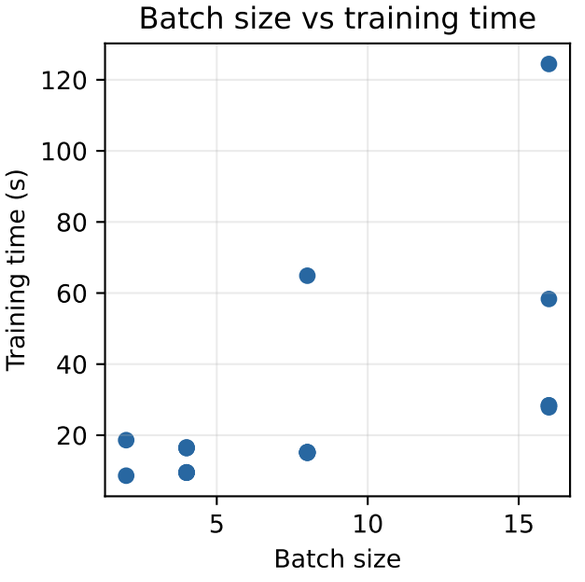


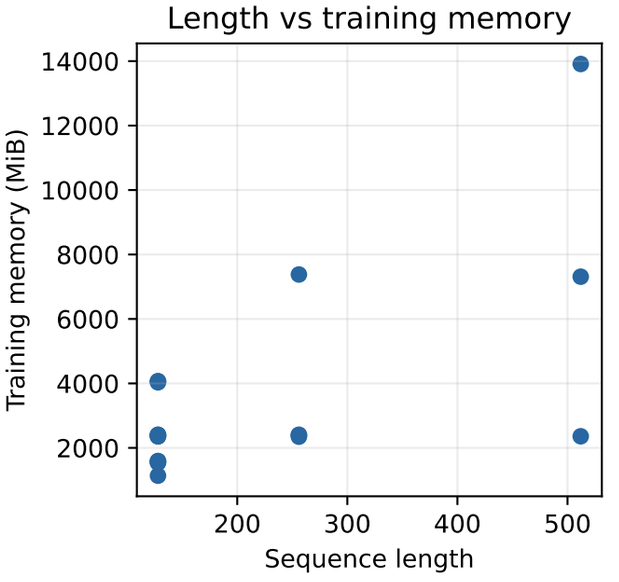












parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention): trial hyperparameters

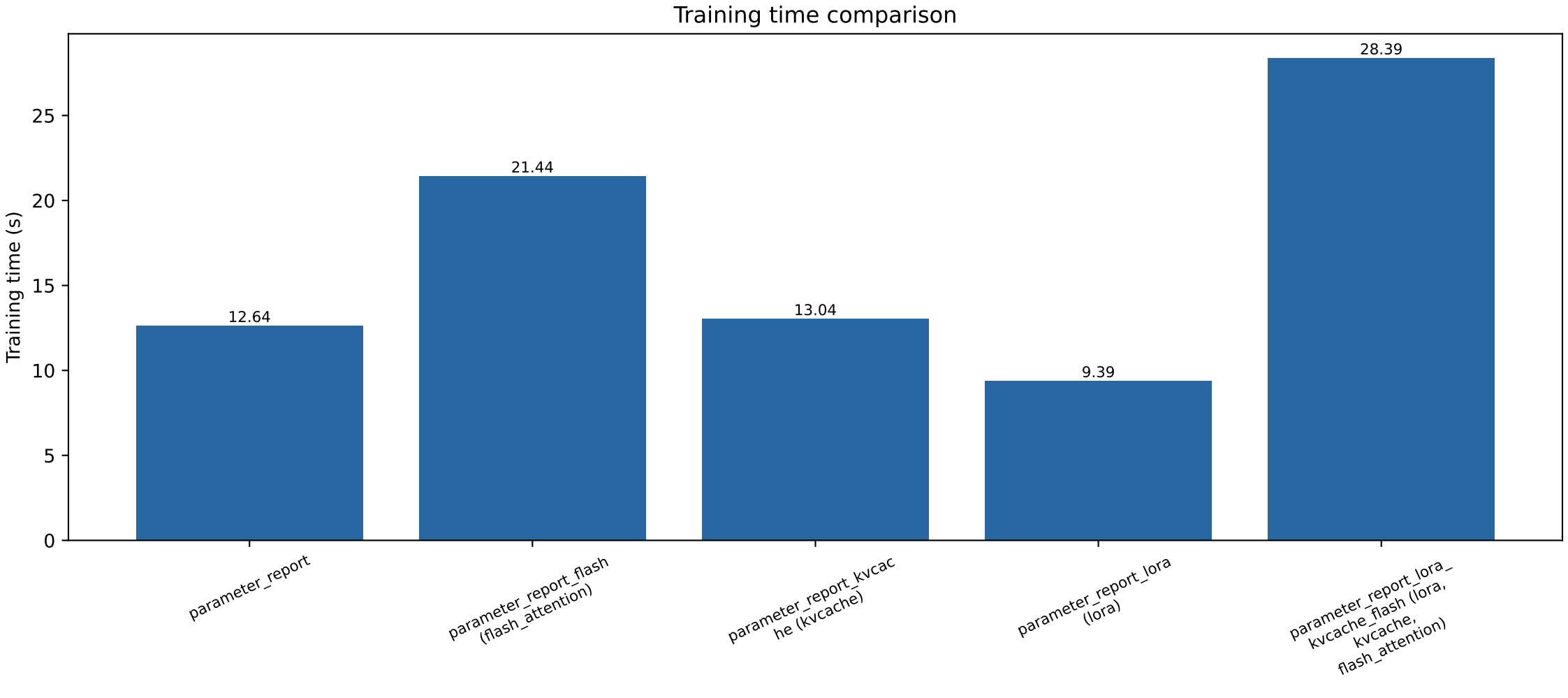
Trial	Learning Rate	Batch Size	Length	Rank	Alpha
0	0.0029578	2	512	8	16
1	9.7448e-05	8	128	8	16
2	2.807e-05	4	256	2	8
3	0.00059155	16	128	16	32
4	0.00023533	4	256	32	64
5	1.5861e-05	16	512	2	4
6	7.9071e-05	2	128	16	8
7	0.00081909	8	512	4	32
8	0.0012308	4	128	2	8
9	4.2976e-05	16	256	32	32
10	0.00059421	8	128	8	32
11	0.00049066	4	128	16	32
12	**0.0008851**	**16**	**128**	**32**	**32**
13	0.0017835	16	128	32	32
14	0.00029194	8	128	32	16
15	0.00067921	8	128	32	64
16	2.0726e-05	4	128	32	64
17	5.8222e-05	8	128	32	64
18	0.0021417	4	256	32	64
19	0.00092976	4	128	32	64

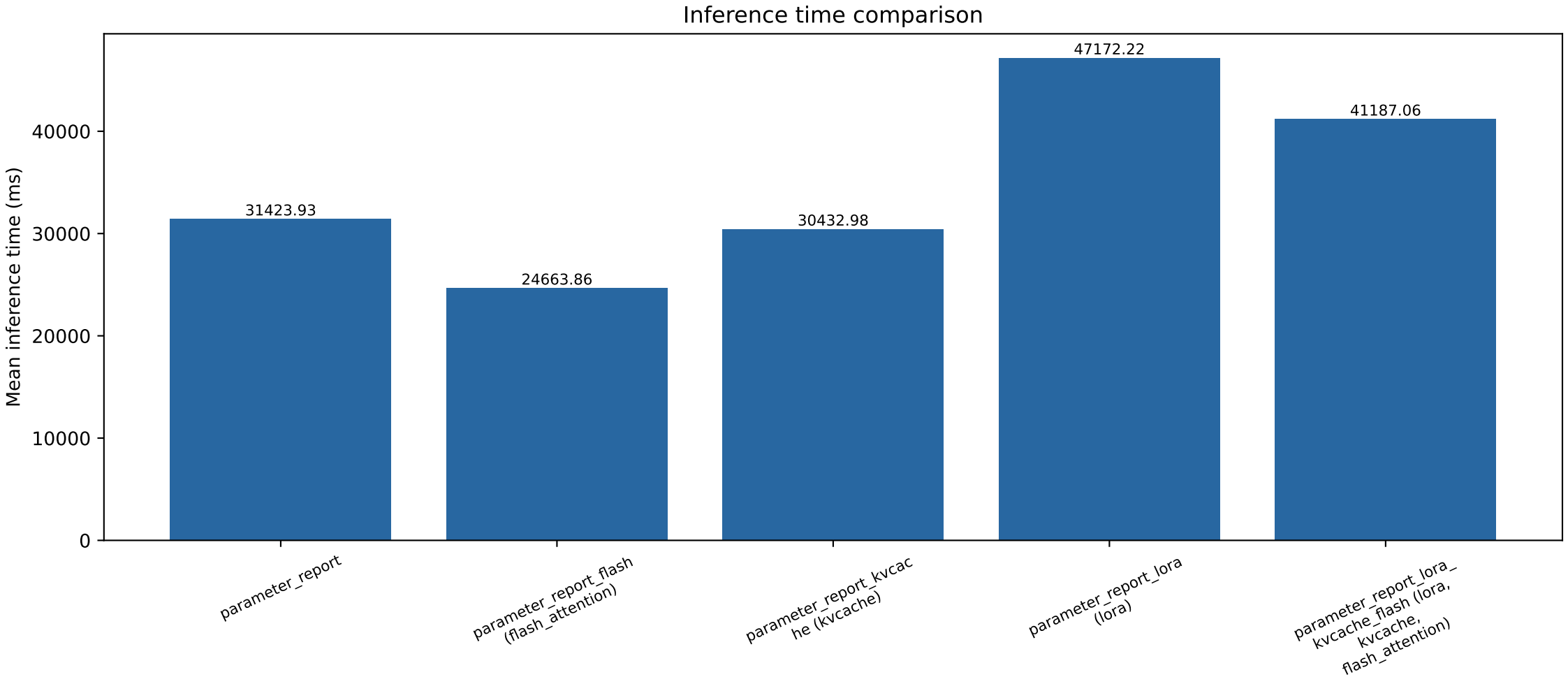
Highlighted row: selected best trial based on this report's objectives.

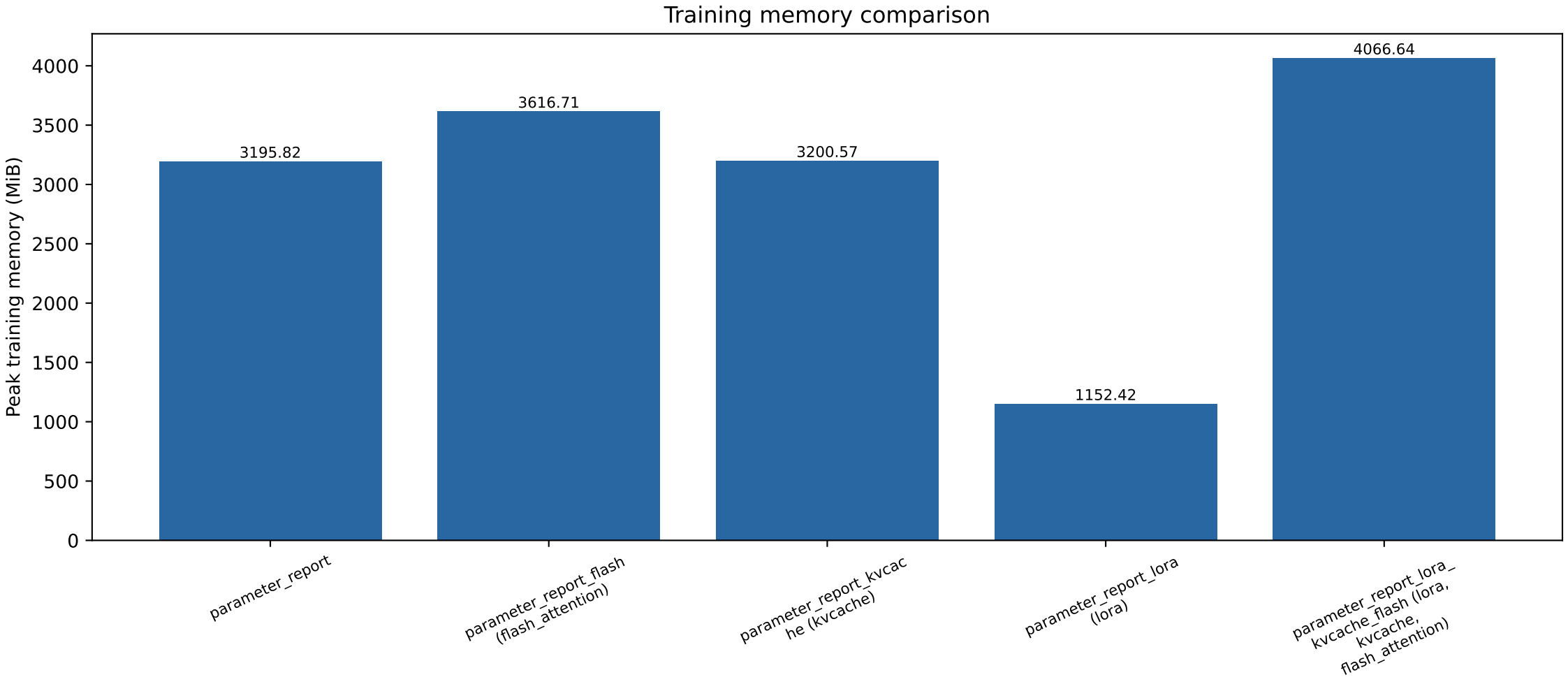
Best trial selected from each parameter-search report

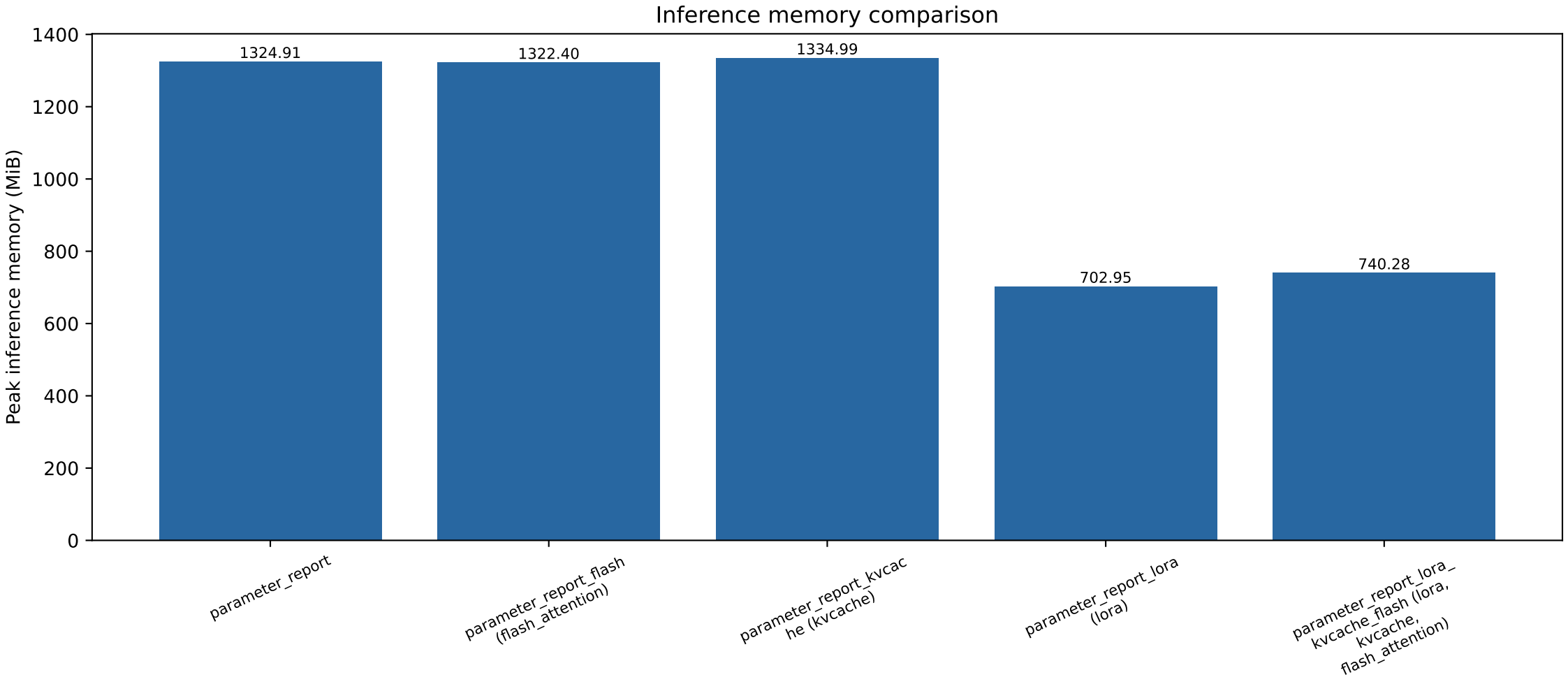
Configuration	Trial	LR	Batch	Length	Rank	Alpha	Train s	Train MiB	Infer ms	Val loss	Val fields	Test fields
parameter_report	4	0.0002262	4	128	-	-	12.6	3196	31423.9	0.1488	50.8%	15.8%
parameter_report_flash (flash_attention)	2	0.0001554	4	256	-	-	21.4	3617	24663.9	0.1391	52.5%	16.7%
parameter_report_kvcache (kvcache)	18	3.574e-05	2	256	-	-	13.0	3201	30433.0	0.1144	60.0%	40.8%
parameter_report_lora (lora)	12	0.001573	2	128	2	8.0	9.4	1152	47172.2	0.1492	49.2%	42.5%
parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention)	12	0.0008851	16	128	32	32.0	28.4	4067	41187.1	0.07074	60.0%	60.0%

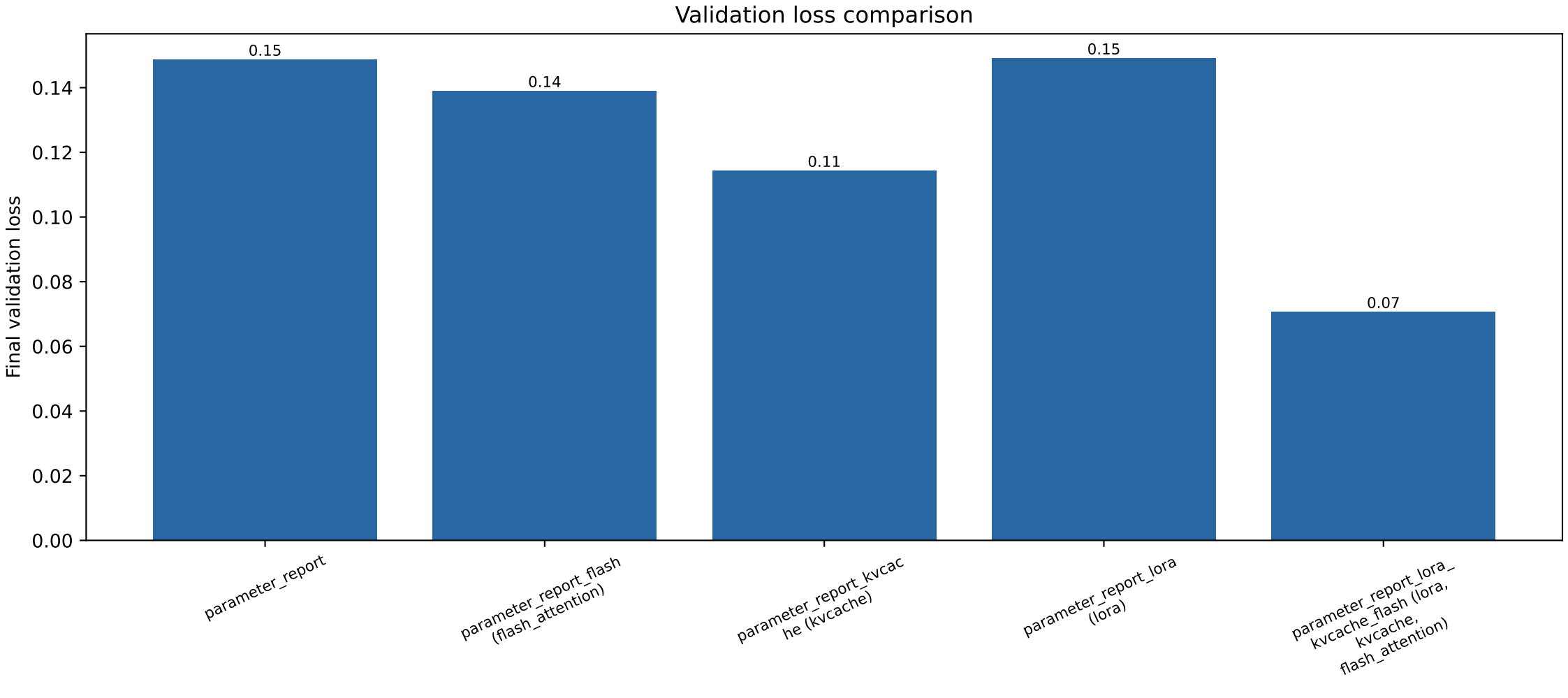
Rows are the trials Ax selected as Pareto-optimal; no trial was selected by this report builder.

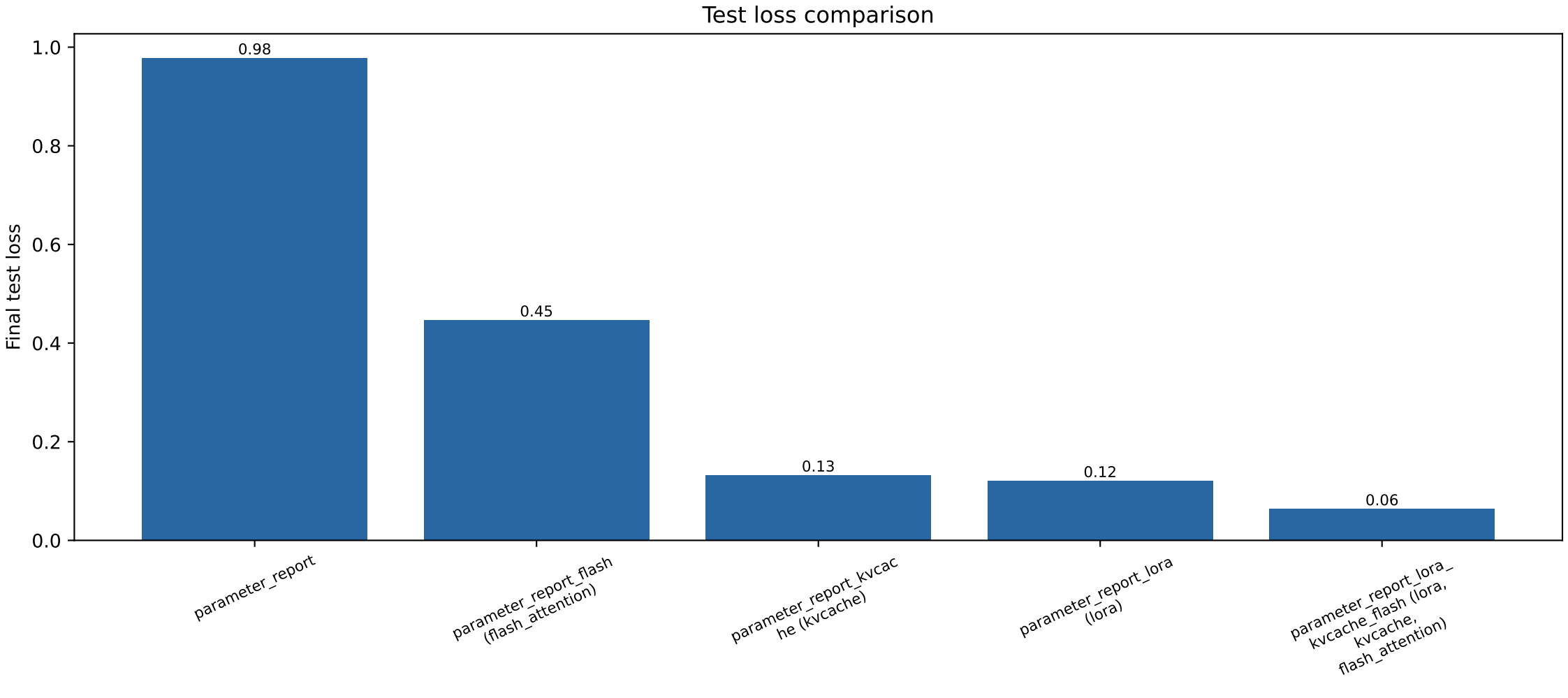

Training time comparison
Training time (s)
0
5
10
15
20
25
12.64
21.44
13.04
9.39
28.39
parameter_report
parameter_report_flash (flash_attention)
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)
parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention)

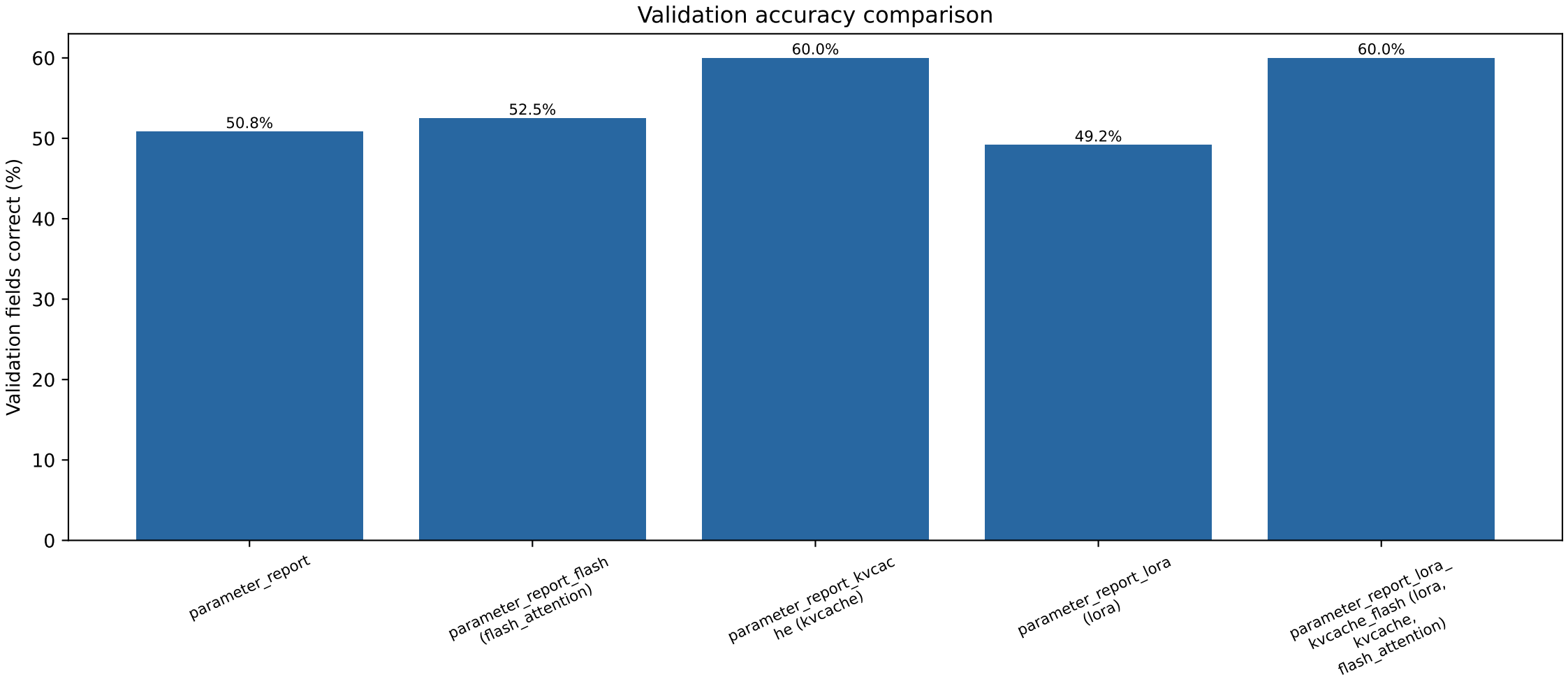

Inference time comparison
Mean inference time (ms)
0
10000
20000
30000
40000
31423.93
24663.86
30432.98
47172.22
41187.06
parameter_report
parameter_report_flash (flash_attention)
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)
parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention)

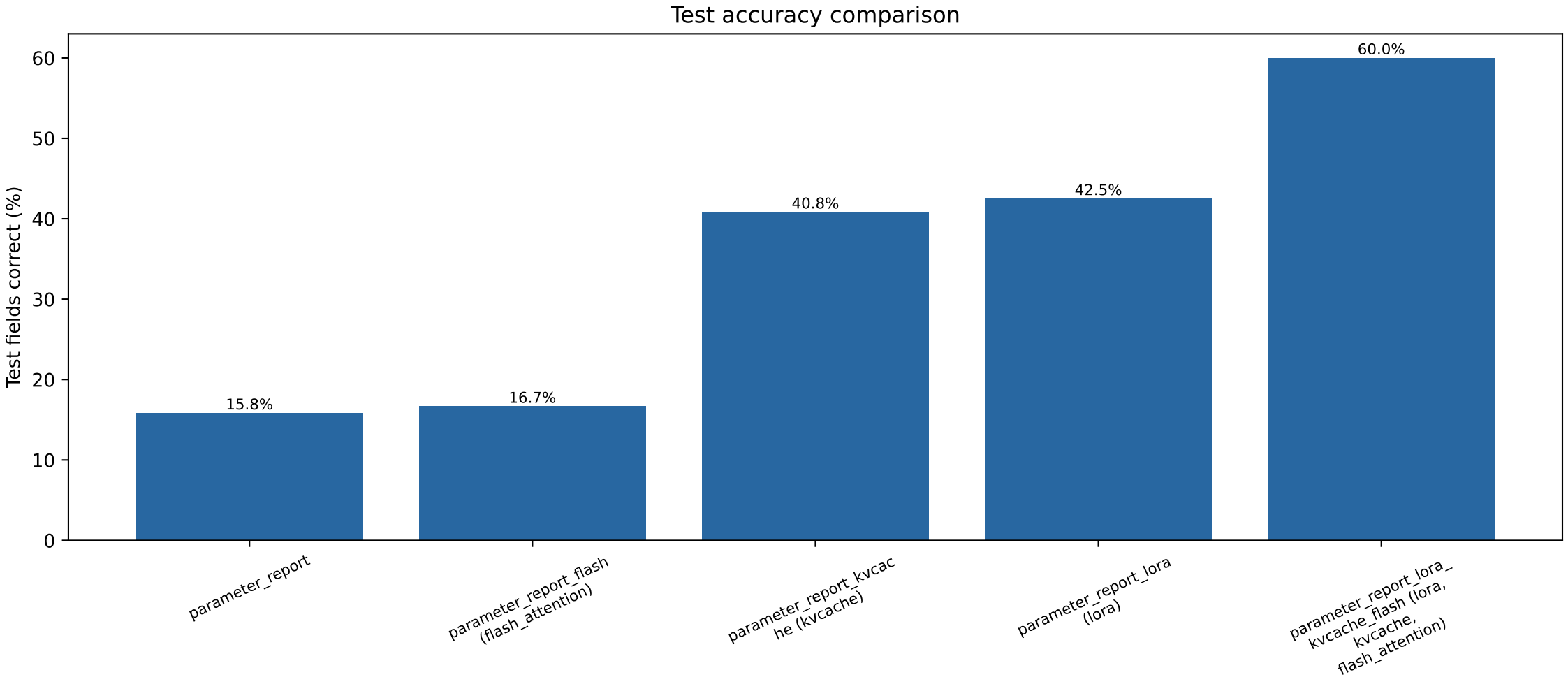

Training memory comparison
Peak training memory (MiB)
0
500
1000
1500
2000
2500
3000
3500
4000
3195.82
3616.71
3200.57
1152.42
4066.64
parameter_report
parameter_report_flash (flash_attention)
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)
parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention)


Inference memory comparison
Peak inference memory (MiB)
0
200
400
600
800
1000
1200
1400
1324.91
1322.40
1334.99
702.95
740.28
parameter_report
parameter_report_flash (flash_attention)
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)
parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention)


Validation loss comparison
Final validation loss
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.15
0.14
0.11
0.15
0.07
parameter_report
parameter_report_flash (flash_attention)
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)
parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention)


Test loss comparison
Final test loss
1.0
0.8
0.6
0.4
0.2
0.0
0.98
0.45
0.13
0.12
0.06
parameter_report
parameter_report_flash (flash_attention)
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)
parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention)


Validation accuracy comparison
Validation fields correct (%)
0
10
20
30
40
50
60
50.8%
52.5%
60.0%
49.2%
60.0%
parameter_report
parameter_report_flash (flash_attention)
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)
parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention)


Test accuracy comparison
Test fields correct (%)
0
10
20
30
40
50
60
15.8%
16.7%
40.8%
42.5%
60.0%
parameter_report
parameter_report_flash (flash_attention)
parameter_report_kvcache (kvcache)
parameter_report_lora (lora)
parameter_report_lora_kvcache_flash (lora, kvcache, flash_attention)